

Solar-Powered Agarbatti Drying & Packaging System — MVP Design Document

Problem Statement: SIH26022 — Solar-Powered Drying and Packaging System for Home-Based Agarbatti Manufacturing by Rural Women Artisans (Ministry of MSME)

Context: Internal hackathon MVP, built to qualify for Smart India Hackathon 2026. Target batch size: ~50-100 agarbatti sticks per drying cycle.

1. Structural Design

Airflow architecture: Bottom-side air intake, top exhaust, fan-assisted (not passive convection).

Air path: Bottom intake → fan (ambient air only) → heating zone (coil) → diffuser/plenum → Tray 1 → Tray 2 → top exhaust vents.

Fan placement: Single fan, bottom-mounted, positioned *before* the heating element in the airflow path — so the fan only ever handles cool ambient air, not heated air. This protects a standard-rated fan from thermal stress and avoids needing a second fan at the top (a single well-placed fan is sufficient for this chamber size; two fans risk fighting each other on pressure balance).

Diffuser/plenum: A perforated plate sits between the heating coil and the first tray, evening out airflow so the bottom tray doesn't get a direct, uneven blast of hot air while the top tray runs cooler. This matters for uniform drying — uneven drying risks cracking/warping in the sticks.

Trays: 2 trays, aluminum wire crisscross mesh, sized for the ~50-100 stick MVP batch. Rough tray size ~20cm × 25cm each, wire gauge 2-3mm (stiff enough to avoid sagging over a 10-15hr cycle).

Rough chamber dimensions:

- Tray spacing: 5-6cm vertical gap (for airflow around sticks)
 - Plenum height below Tray 1: 8-10cm (room for the diffuser to actually even out flow)
 - Overall internal chamber height: ~40-45cm (tray stack + plenum + headroom for sensors/exhaust)
 - Chamber footprint: ~25cm × 30cm (wall clearance around trays)
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Temperature target: ~60°C, chosen as the working hypothesis for safe agarbatti drying (avoids case-hardening/cracking risk from flash-drying, stays below where most fragrance/essential oil compounds start degrading). Treated as a hypothesis to validate with a real test batch (checking for cracking, fragrance loss, breakage rate), not a confirmed-optimal figure — no hard literature source was found confirming this exact number is optimal.

2. Materials

Material choice driven by: 60°C operating target, with a 100-120°C sustained-tolerance safety margin (10-15hr runs) for summer ambient conditions and fault scenarios (e.g., thermostat lag, coil hot-spotting).

Component	Material	Spec	Reasoning
Inner chamber walls	Aluminum sheet	0.8-1mm	Handles 100-120°C with large margin, easy to cut/rivet at small scale, light and portable
Diffuser plate	Perforated aluminum sheet	Same stock, ~5-6mm holes, 40-50% open area	Same material as walls — simplifies sourcing
Trays	Aluminum wire mesh, crisscross	2-3mm wire gauge	Max airflow, easy to clean, resists sag at this scale
Outer housing	Plywood/MDF	9-12mm	Outer shell doesn't need heat tolerance if inner shell + air gap does the thermal work; cheap, easy to hand-fabricate
Inner-outer air gap	Air gap (insulation optional)	1.5-2cm	Reduces heat transfer to outer shell and ambient, improves efficiency and touch-safety

Alternative outer shell option (cost-reduction idea): Repurposing a used 15L edible oil tin ("tel ka pipa") as the *outer* housing instead of plywood — these are cheap/free and

locally available. **Ruled out for the inner (heat-racing) chamber** specifically, due to risk of interior lacquer/lining off-gassing or degrading under sustained heat, which could contaminate fragrance/product — a real risk given the product's whole value is its aroma, not just a "might be fine."

Rough structural material cost: ~₹1,200-2,000 total (aluminum sheet + mesh + plywood + fasteners), sourced locally (Bhilai/Durg hardware/sheet-metal shops rather than online bulk suppliers, since MVP quantities are too small for typical B2B minimum orders). Worth checking the college workshop/mechanical lab for offcuts before buying.

3. Power System

Design: Solar PV + battery as primary power, grid mains as backup when battery is depleted or sunlight is insufficient. Deliberately **not** solar thermal (too slow/unregulated for precise temperature control) and **not** 100% off-grid (unrealistic reliability for an artisan who needs the machine to always work).

Available hardware: A Tata BP Solar TBP1275M panel (75W, 17V Vmp, 4.4A Imp, 21.8V Voc) is already owned and will be used for the MVP.

MVP power decision: For the actual internal-hackathon demo, use **direct mains-adapter/plug supply** as the real power source (simpler, no battery cost/complexity, more reliable for a live demo). The solar panel is wired in to visibly demonstrate the concept (e.g., contributing to or charging a small buffer). **Battery sizing/purchase is deferred** — not built for this MVP, explicitly because of hackathon budget/timeline scope, not because it's unnecessary. If the team advances to the full SIH round, battery backup will actually be built and invested in.

4. Heating & Fan Sizing

Heating coil: No confident wattage number exists yet — industrial-scale reference data doesn't scale down reliably to this chamber size. **Plan: build the empty chamber shell first (walls + plenum, no trays/sticks needed), start with a modest ~150-200W coil, and empirically test** — log temperature rise over time using the DS18B20 sensors, then adjust coil size up or down based on real data. This is faster and more accurate than calculating from a mismatched industrial reference.

Fan: Less critical to test precisely. A small 12V DC fan (80-92mm computer-style, ~20-40 CFM) is a safe, low-risk starting point for this chamber size. Main risk is over-blowing

(sticks moved, uneven surface drying), correctable via PWM speed control if needed.

5. Sensing

Temperature: 2× DS18B20 digital sensors (chosen over analog NTC thermistors for simpler, more reliable digital readout, less calibration overhead, multiple sensors shareable on one data pin).

- Sensor 1: plenum-side (near the coil, for fast control-loop response)
- Sensor 2: tray-side (near the trays, to confirm actual drying-zone temperature — this is what matters for the product, not the coil's local reading)

Humidity: A humidity sensor (DHT22 — chosen over DHT11 for wider temperature/humidity tolerance) placed inside the chamber, used for moisture/completion sensing rather than running a purely fixed-time cycle.

Completion logic: Plateau-detection, not a fixed threshold — track humidity trend over a rolling time window; when the drop in humidity flattens out (diminishing returns), mark the cycle complete. A fixed absolute humidity threshold would be unreliable since ambient humidity varies day to day.

6. Safety Design (defense in depth — three independent layers)

1. **Software control loop:** Main thermostat-style logic — sensor reads temperature, relay/MOSFET toggles the coil to hold ~58-62°C.
 2. **Independent firmware overheat cutoff:** A hardcoded check, separate from the main control loop — if any sensor reads above a safety threshold (e.g., >90°C), force-cut coil power regardless of what the main loop is doing. This protects against bugs or logic errors in the primary control code.
 3. **Physical thermal fuse:** A one-time, non-resettable thermal cutoff (rated e.g. 130-150°C) wired directly in series with the coil's power line — a purely hardware, dumb backstop that works even if the MCU, sensors, or firmware fail entirely. User must replace it if it trips, which is a deliberate design choice (signals "something went wrong, get it checked" rather than silently auto-resetting).
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7. Firmware / Control Architecture

MCU: ESP32 (chosen over Arduino Uno for built-in WiFi — enables IoT monitoring/alerts as named in the PS — plus more GPIO/processing headroom for the sensor+relay+display combination).

Control logic: State machine, not a flat loop.

IDLE

→ user starts cycle → HEATING

HEATING

- Fan ON
- Coil ON if temp < 60°C, OFF if >= 60°C
- → MAINTAINING once at target
- [always] hard cutoff → FAULT if any sensor > 90°C

MAINTAINING

- Fan ON, coil cycles to hold ~58-62°C band
- Humidity logged on rolling window; plateau detected → COMPLETE
- [always] overheat cutoff active

COMPLETE

- Coil OFF
- Fan stays ON for a fixed cooldown period (safety: avoids opening a hot chamber)
- → READY_TO_OPEN once cooldown finishes, signal "safe to open"

FAULT

- Coil OFF (latched, requires manual reset)
- Fan can stay ON to help cool the chamber
- Fault clearly signaled

Cooldown approach: Fixed-time cooldown (not temperature-threshold) — simpler to implement and more predictable for a live demo.

Component/pin mapping: | Component | Interface | |---|---| | DS18B20 ×2 | OneWire, shared bus, 1 GPIO + 4.7kΩ pull-up | | DHT22 | Single-wire digital, 1 GPIO | | Heating coil | Relay/SSR (never direct GPIO — coil current exceeds GPIO limits) | | Fan | MOSFET (PWM if variable speed wanted) | | Status LEDs | 3× GPIO with current-limiting resistors | | Optional display | I2C OLED/LCD (SDA/SCL) | | Thermal fuse | Hardware-only, in series with coil's mains-side wiring — not an MCU pin |

Safety note: the relay/SSR and thermal fuse sit on the mains/high-current side of the circuit — this is the one part of the build where a mistake has real consequences, worth having an experienced person double-check before first power-on.

8. Packaging Station

Context: Target market sells agarbatti by weight (e.g., 100g packets), not stick count — this decided the sensing approach.

Baseline mechanism: Manual poly bag/pouch + lever-operated impulse sealer.

Added sensing element: Load cell + HX711 amplifier for weight-based portioning.

Flow:

1. Artisan places empty bag on load cell platform, tares/zeroes
2. Artisan manually adds sticks
3. System reads weight continuously
4. Near target (e.g., 90g of 100g) → slow-down signal
5. At target weight ($\pm$ tolerance) → "done" signal
6. Artisan removes bag, seals manually

Count-based alternative (documented fallback, not built): If market research later shows count-based selling instead, an IR break-beam sensor + a narrow singulating chute/funnel could count sticks one at a time as an alternate mode. Noted as a pivot option in the pitch, not built for this MVP, since actual target market confirmed weight-based selling.

Explicitly out of scope for MVP (documented as "future roadmap" in the pitch):

- Automated bag advance/dispense mechanism
- Automated sealing trigger (motorized/solenoid)
- Any conveyor/feeding automation
- Label/date stamping automation

Rough packaging station cost: ~₹650-1,250 (load cell + HX711 + impulse sealer + bags + mounting hardware).

9. Packaging Material

PS requirement: Low-cost, moisture-resistant, aroma-preserving, eco-friendly packaging suitable for SHG/women artisans — this is final consumer-facing retail packaging (the actual sealed 100g packet reaching a buyer), not intermediate bulk storage.

Real tradeoff acknowledged: the most eco-friendly materials (plain kraft paper) have the weakest moisture barrier; the strongest moisture/aroma barriers (BOPP/plastic) aren't eco-friendly. No single material perfectly satisfies both.

Pitch position:

- **Primary/practical MVP choice:** Plastic pouches (BOPP or HDPE) — proven, cheap, strong moisture/aroma barrier, what most real agarbatti products currently use.
- **Future eco-upgrade path (named in pitch, not built):** Kraft paper pouches with a thin liner (printed/glossy) — better eco-story, still gives moisture protection via the liner, sourceable locally (found a supplier based in Indore). Positioned as a roadmap item as sourcing/cost improves at scale, rather than an MVP overclaim.

This gives an honest answer to the PS's eco-friendly requirement without silently ignoring it or pretending a single material solves the whole tradeoff.

10. Fragrance-Preservation (PS requirement addressed)

Understanding reached (industry-consistent pattern, not explicitly confirmed by a single authoritative source): the dominant agarbatti manufacturing workflow treats "raw incense stick" (rolled, dried, unscented) as a distinct product/stage, separate from "perfume compound"/fragrance oil application — supported by the fact that raw sticks and perfume compounds are sold as separate standalone products in the industry. This suggests fragrance is typically applied **after** drying, meaning the drying chamber itself often doesn't need to protect any fragrance at all.

Where masala is pre-scented (some smaller/traditional producers mix aromatics directly into the paste before rolling): 60°C is well below typical thermal degradation thresholds for essential oils/aromatic compounds, so fragrance loss at this temperature is expected to be minimal — though this specific claim wasn't independently verified against a technical source, and is flagged as a working assumption.

11. Cost Summary

Category	Estimated cost (₹)
Structural materials (aluminum sheet, mesh, plywood, fasteners)	1,200 - 2,000
Electronics (ESP32, sensors, relay/MOSFET,	2 500 - 4 500

heating coil, fan, safety, misc)	~₹1,000 - 1,500
Packaging station (load cell, HX711, sealer, bags, hardware)	650 - 1,250
Combined MVP total (approx.)	~₹3,200 - 5,750 (excluding solar panel, already owned)

Note: These are indicative reference prices based on wholesale/bulk listings and general India market rates as of Aug 2026 — actual local Bhilai/Durg pricing may vary. Confirm locally before finalizing budget.

Open Items / Next Steps

- Empirically test heating coil wattage on the built chamber shell
- Validate 60°C drying assumption with a real test batch (check cracking, fragrance, breakage vs. traditional drying)
- Confirm interior lining type if reconsidering any repurposed containers
- Firmware implementation (Arduino/ESP32 sketch) — not yet written
- Pitch deck / presentation structure — not yet started
- Build sequencing across remaining hackathon days — not yet planned